Distributed Computing: From Edge to Exascale

Architectures for Intelligent Systems in the Real World

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ASPIRE
NSF Engineering Research Center

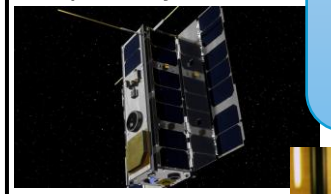
DIRECT LAB

The DIRECT Lab integrates AI, robotics, and electrification to build intelligent, connected infrastructure. We have deployed solutions in space, ground robots, and large-scale transportation systems.

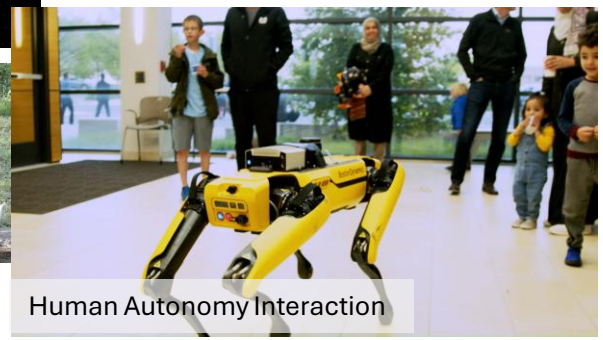
Mario Harper

Autonomy and Robotics

Space Systems

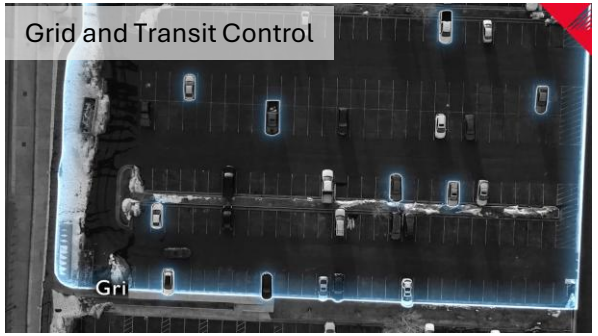


Ground Robots



Human Autonomy Interaction

Electrified Mobility



AI Systems and Digital Twins

Long-Range Mission Simulator

Multiagent Robotics Simulation

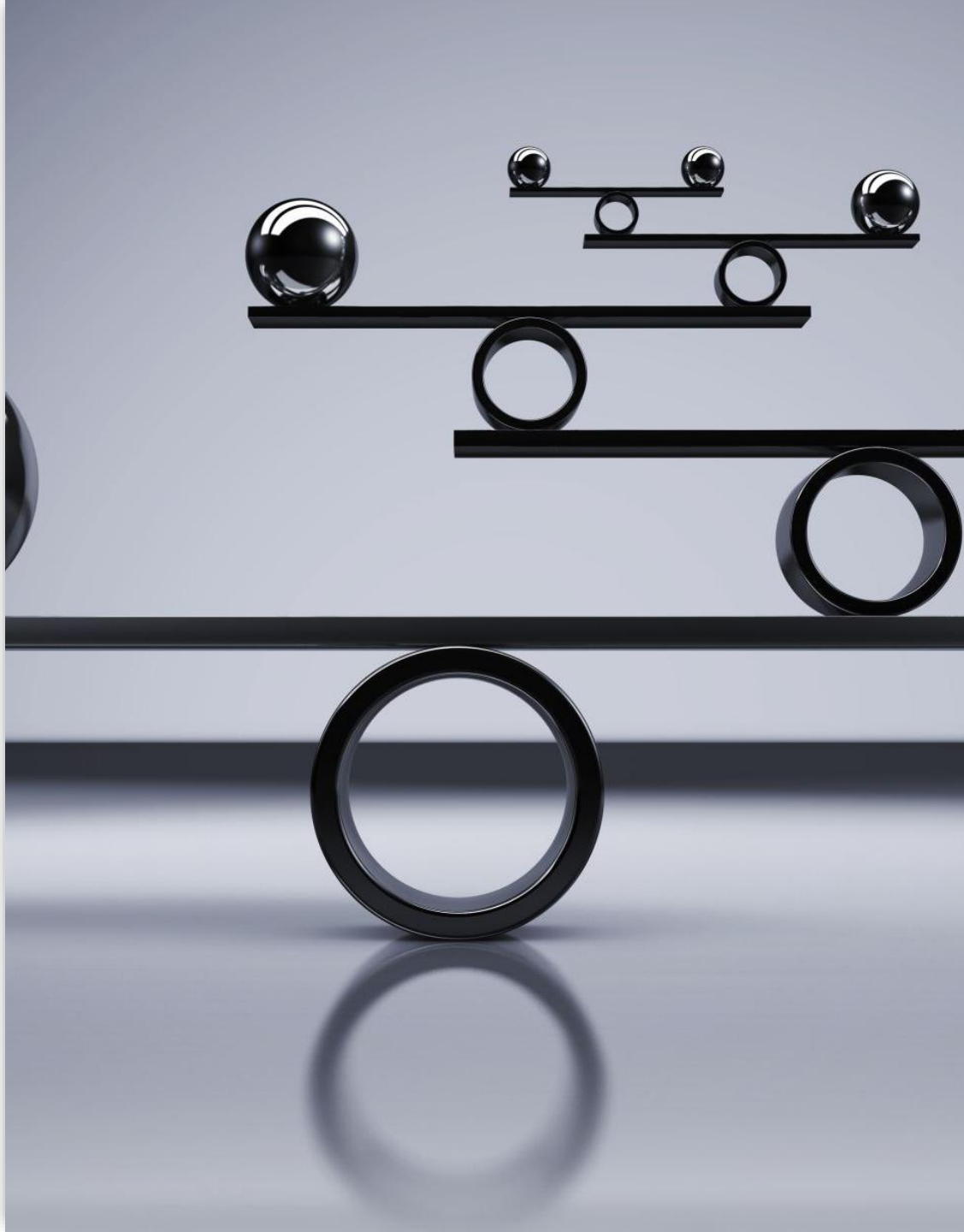
Core Partners

NVIDIA · U.S. ARMY · DEVCOM ARMY RESEARCH LABORATORY · 309th SOFTWARE ENGINEERING GROUP · U.S. AIR FORCE · AlphaZ · NASA · SPACE DYNAMICS LABORATORY

PACIFIC POWER · UTA · GEOTAB · Adobe · ABB · UTAH CLEAN CITIES · Goldman Sachs · NICE · ASI · raisin · driivz

Why Distributed Computing Exists

- Latency constraints
- Physical distribution of sensors
- Data gravity and bandwidth limits
- Fault tolerance and resilience
- Power constraints and scale



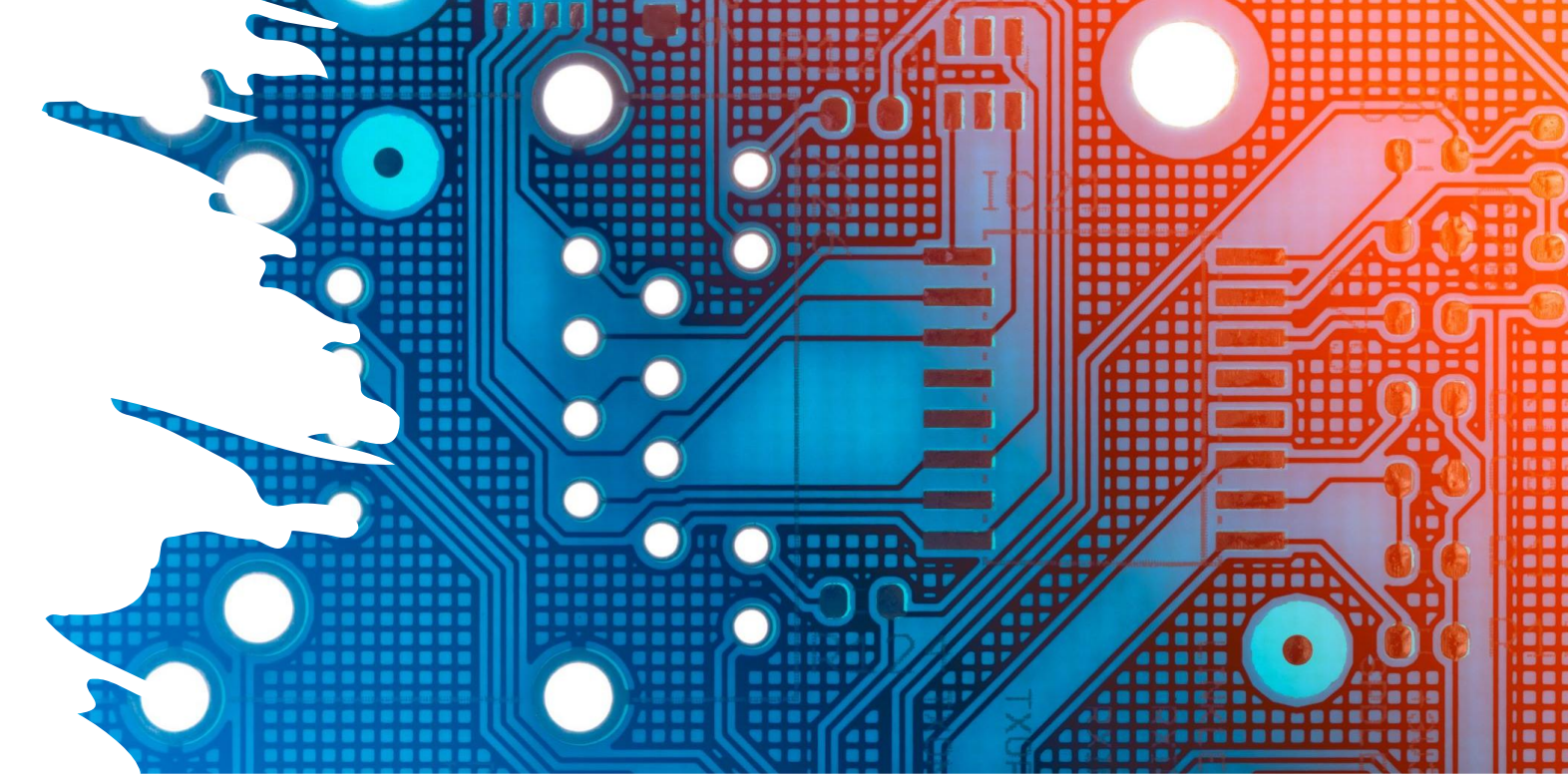
Architecture Spectrum

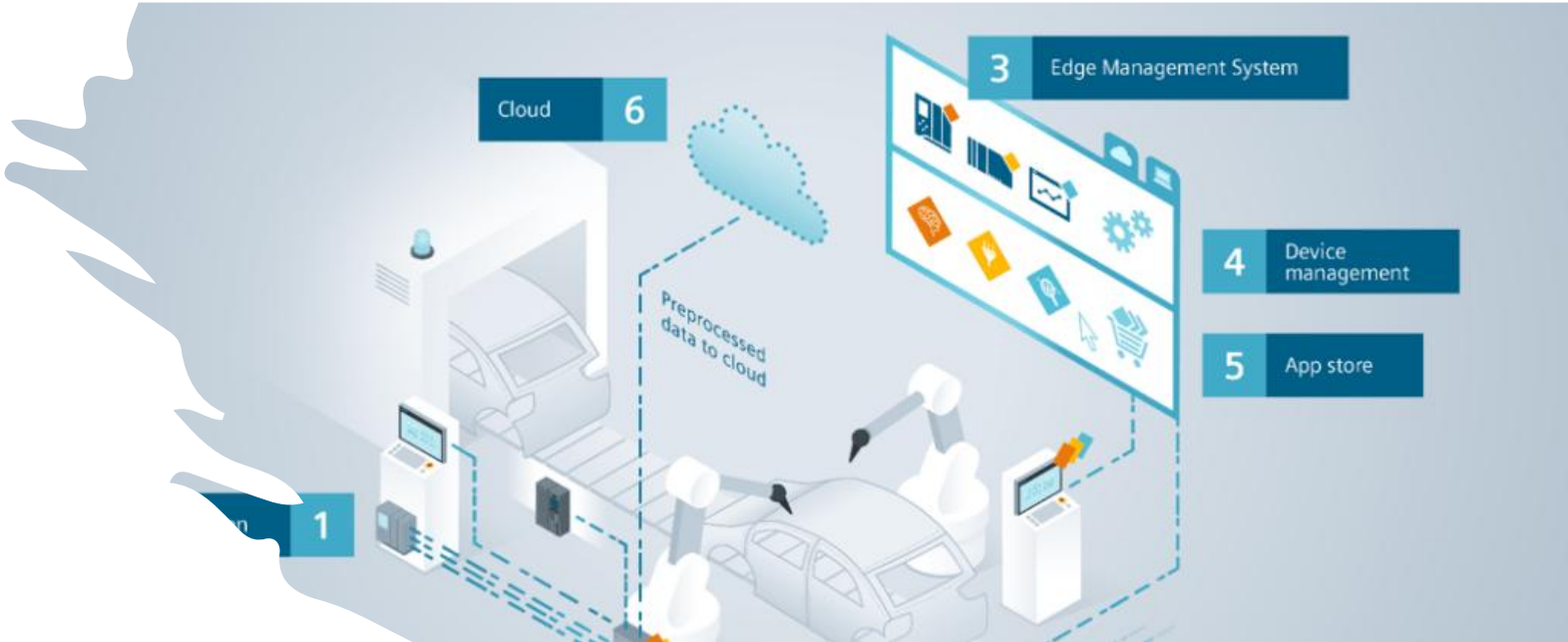
Edge → Federated → HPC → Kubernetes Cloud

Hybrid systems dominate in practice

Edge Computing

- Sensor → Local Compute → Decision → Actuation
- Ultra-low latency
- Must tolerate disconnection
- Power and memory constrained





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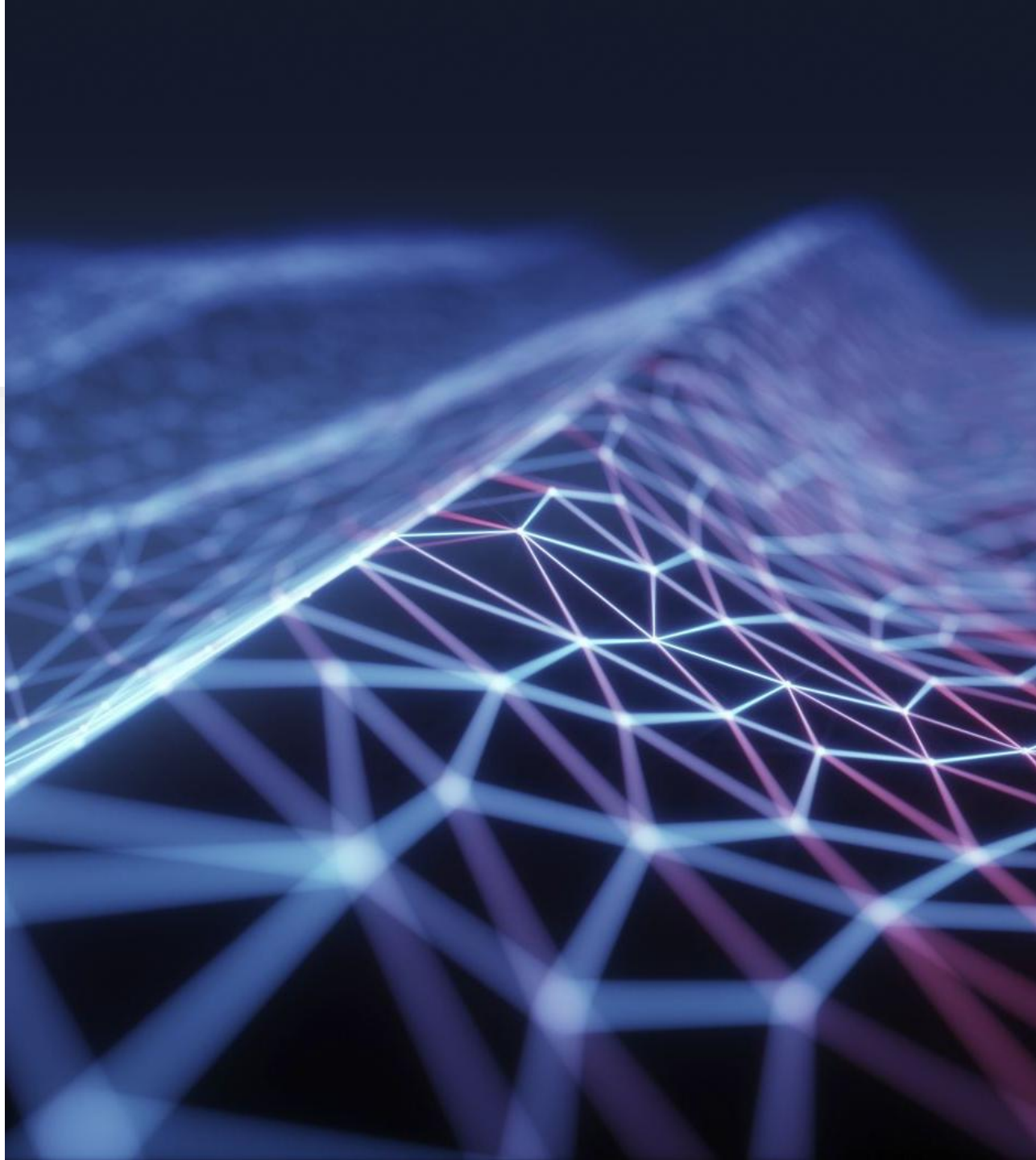
cradlepoint

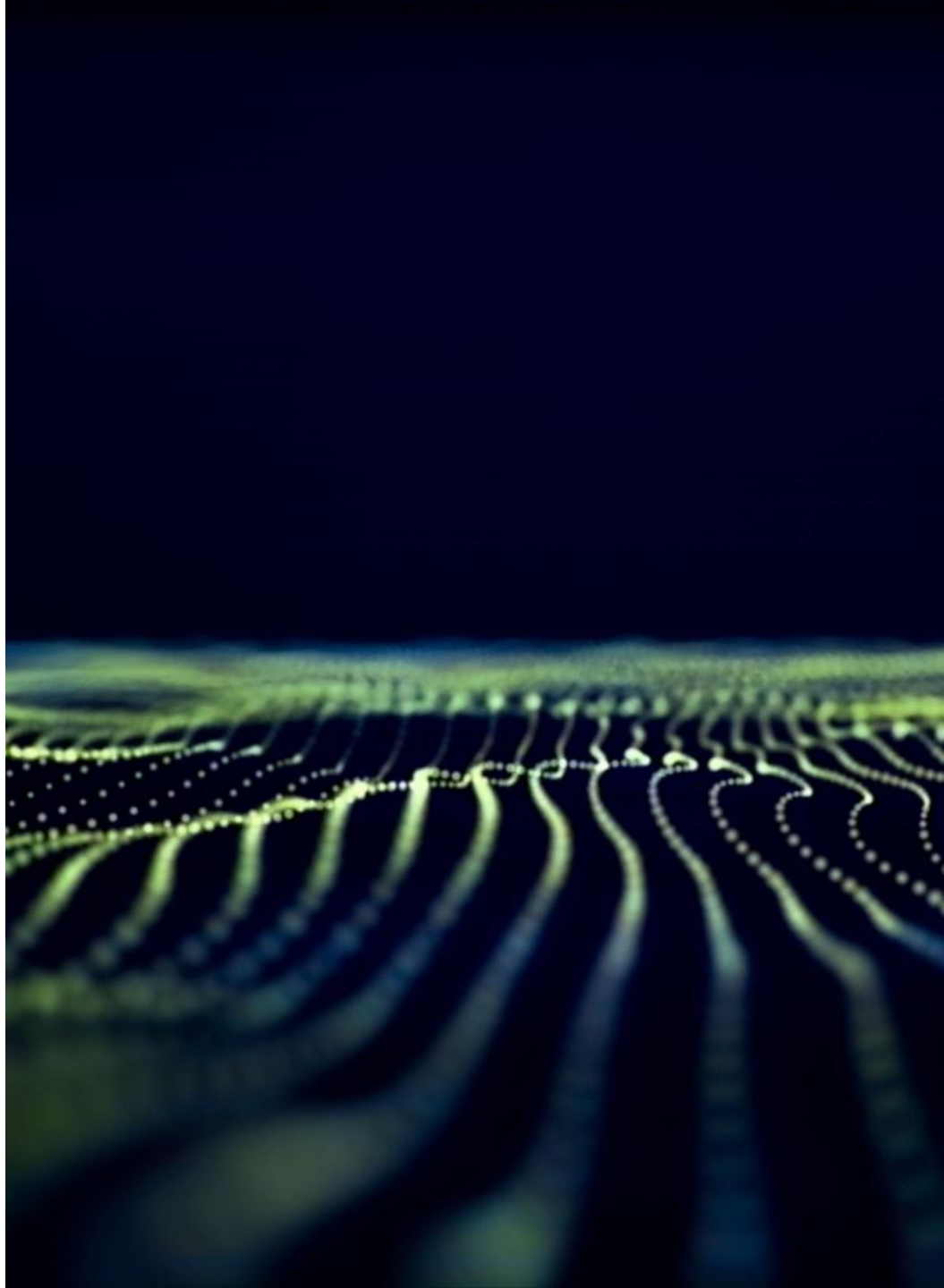
Human-Rich Robotics Example

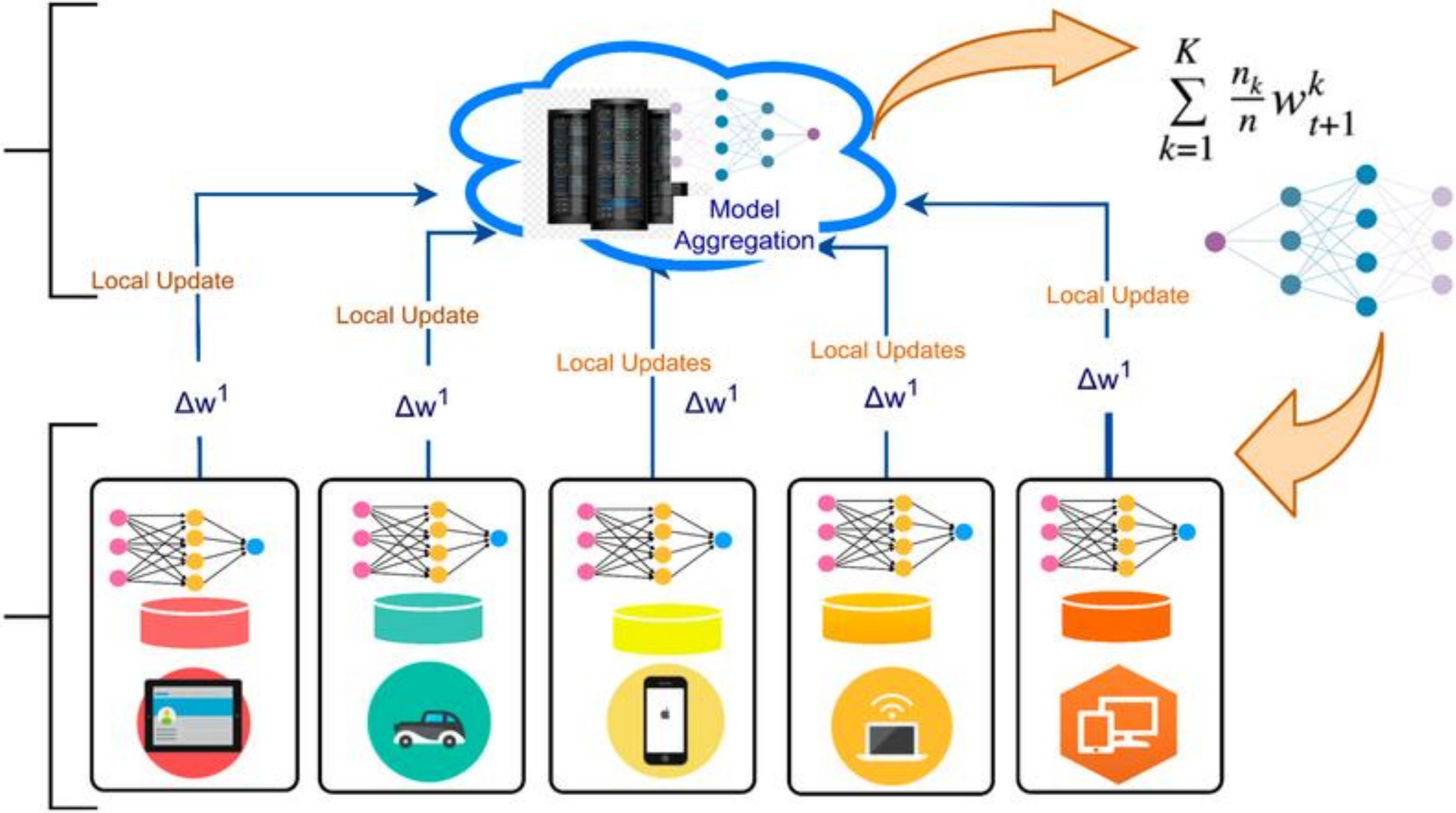
- Dynamic obstacle updates
- Probabilistic occupancy grids
- Receding-horizon replanning
- Continuous sensor fusion



Federated Systems

- Local training
- Gradient aggregation
- Privacy preservation
- Non-IID challenges




$$\sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^{k}$$
Model Aggregation
Local Update
Local Update
Local Updates
Local Updates
Local Update
Δw^1
Δw^1
Δw^1
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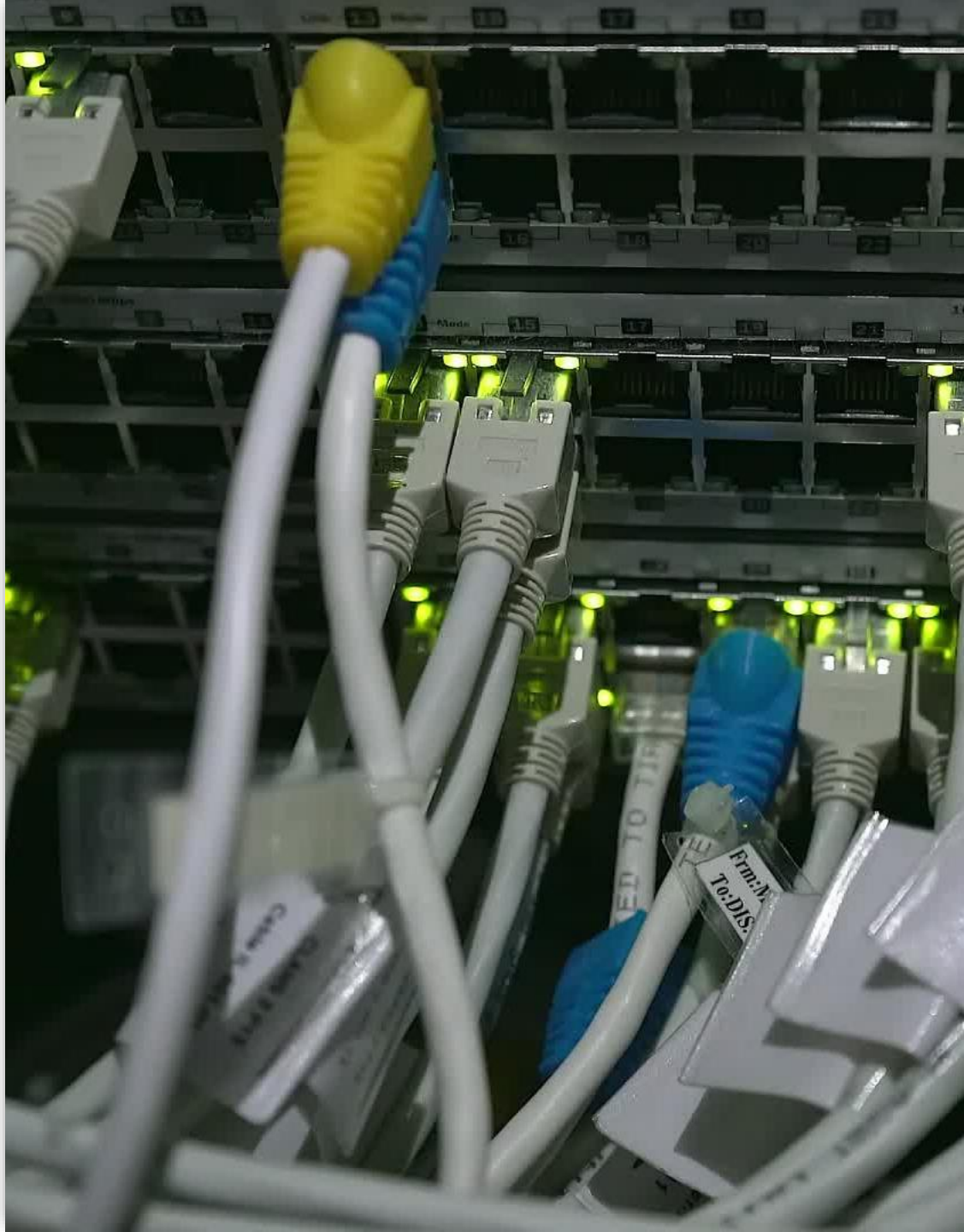
Δw^1

HPC Clusters

- Head node and scheduler
- GPU compute nodes
- Shared storage
- Ideal for training and simulation



SLURM vs Kubernetes

SLURM: BATCH HPC SCHEDULING

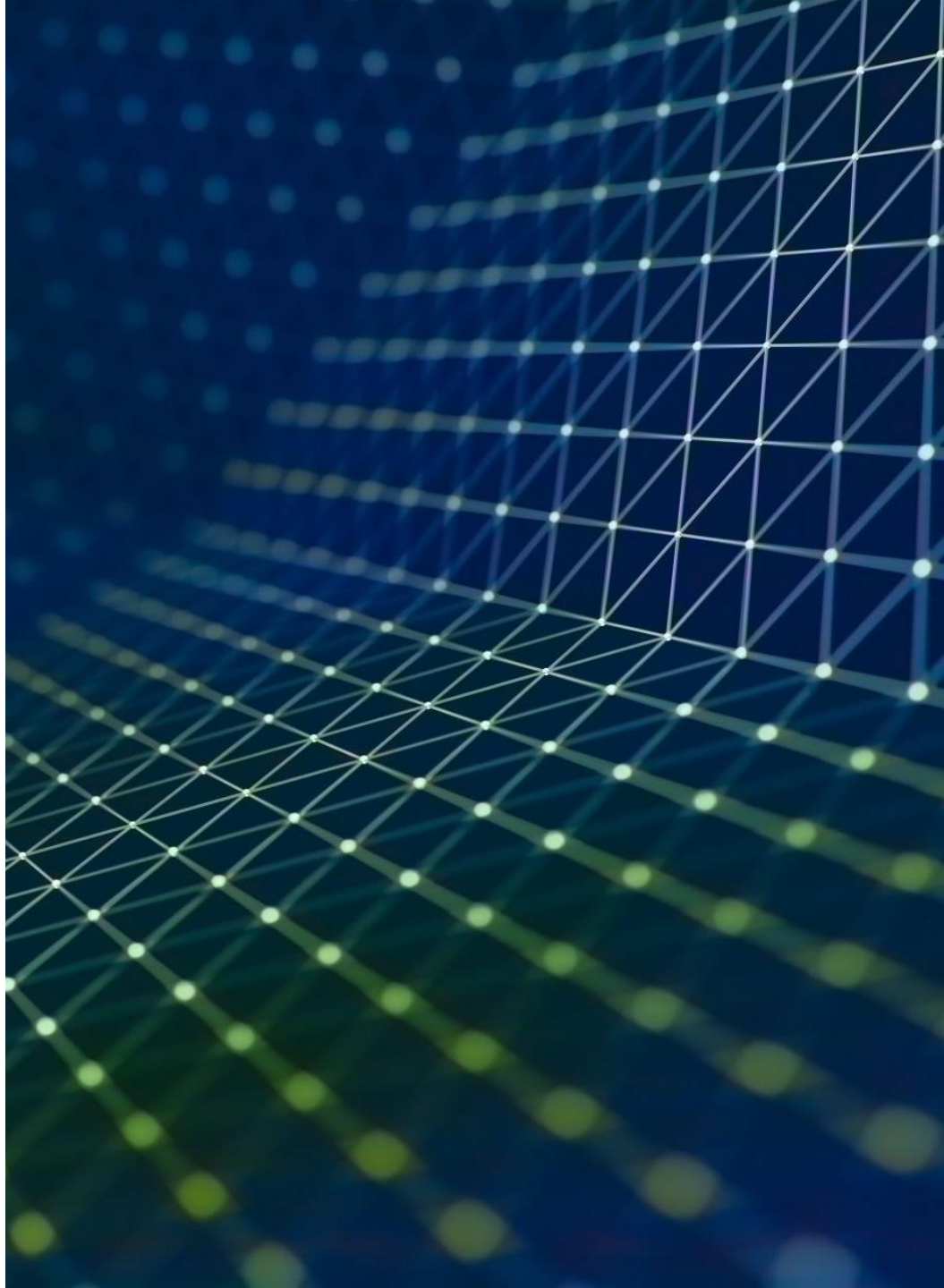
KUBERNETES: SERVICE ORCHESTRATION

BOTH GPU AWARE

DIFFERENT SCALING PHILOSOPHIES

Kubernetes Architecture

- Control plane
- Worker nodes
- Pods and services
- Auto scaling and self-healing



Design Axes in Distributed Systems

Where compute lives

How coordination occurs

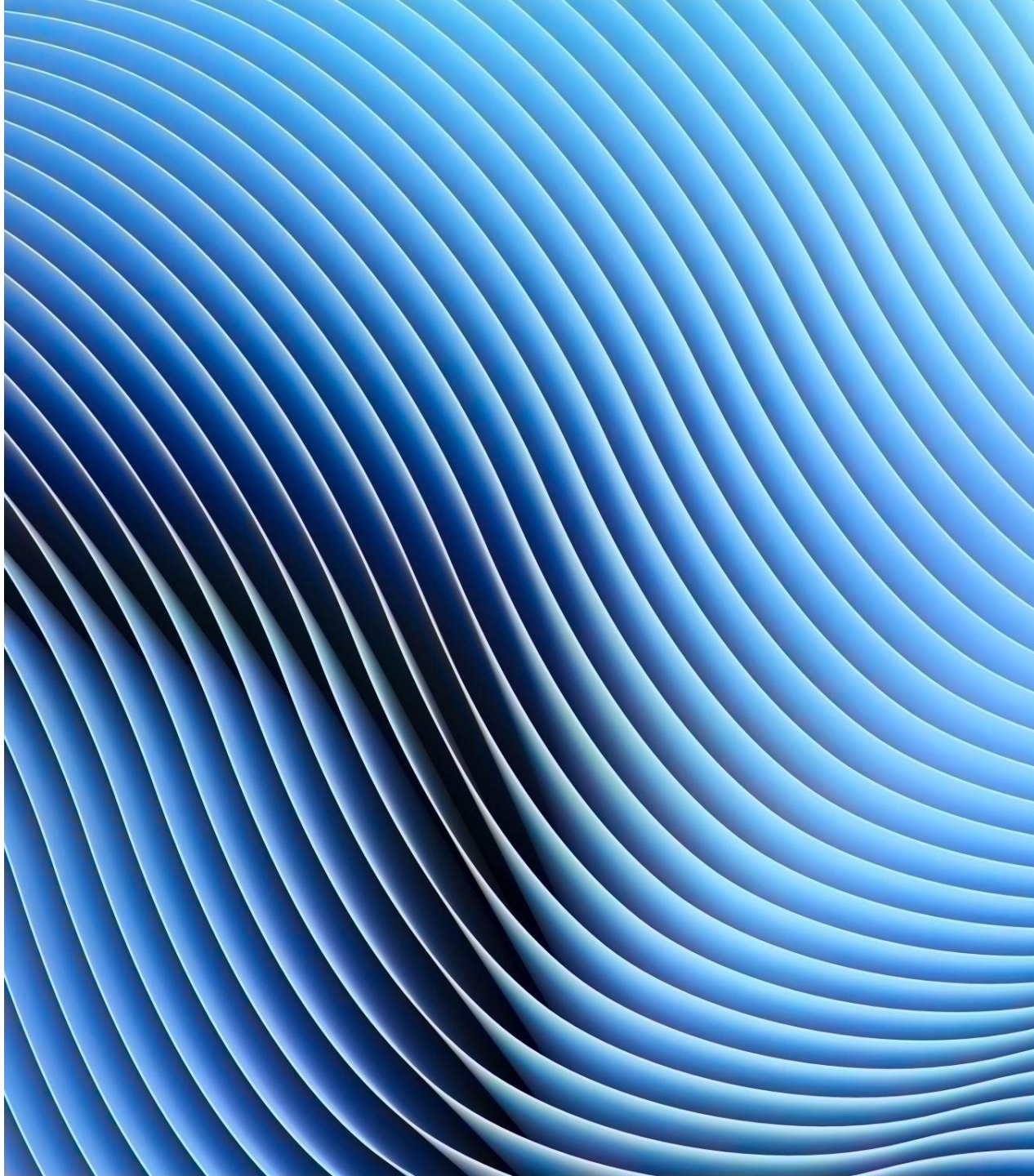
How failure is handled

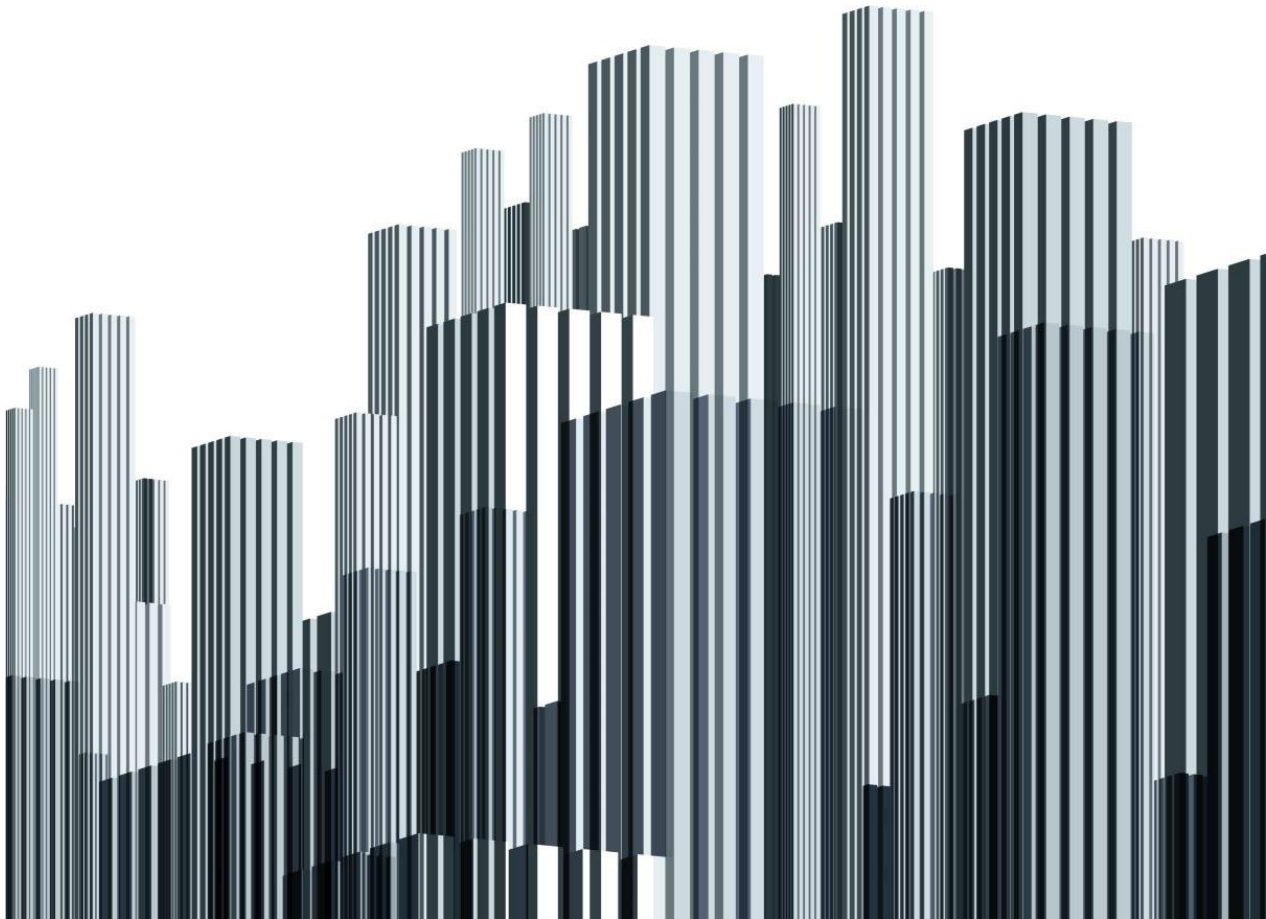
Real-World Example: Autonomous Fleet

- Edge inference onboard vehicles
- Federated updates across fleet
- HPC retraining
- Kubernetes dashboards and APIs

Failure Handling

- Edge failure
- Node failure
- Cluster failure
- Network partition



Closing Thoughts

- Hybrid architectures dominate
- Distributed intelligence is the future
- Design for scale and resilience